

# Probabilistic Machine Learning

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## Lecture 05: Bayesian Networks

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Speaker notes

Belief networks in depth, plus the two topics requested for this lecture: fitting (parameters and structure) and graphical notation. The plate-notation unit is new everything else is reused. Markov networks are included as a contrast — drop that module for a shorter course.



# Module 1

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## Belief Networks

**Guiding Question:** *What exactly does an arrow in a probabilistic model claim — and what does a missing arrow claim?*

# Definition

**Bayesian belief network (BBN).** A **directed acyclic graph** whose nodes are random variables, in which the joint distribution factorises as

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i \mid \text{pa}(x_i))$$

where  $\text{pa}(x_i)$  are the parents of  $x_i$  in the graph.

Each node carries one conditional probability table given its parents. The wet-grass model was exactly this.

Math first (P5) — the students have already built one of these by hand, so the general form should read as a summary rather than a novelty. Stress "acyclic": cycles would make the factorisation ill-defined.

# Three Tasks

## Modelling

Choose variables, edges and tables that encode the domain.

## Inference

Compute posteriors given evidence.  
Exact **and** approximate inference are **NP-hard** in general.

## Learning

Estimate the tables — and possibly the structure — from data.

The NP-hardness is worth dwelling on: tractability depends on the *shape* of the network, not on its size alone. Trees are easy, dense graphs are not.

# Modelling Technique: Divorcing Parents

A node with  $k$  binary parents needs  $2^k$  table entries. **Divorcing** inserts an intermediate variable that summarises a group of parents, so the table becomes two small ones instead of one large one.

The intermediate node usually has a meaning of its own — "total load", "combined risk" — which is why the technique improves interpretability as well as size.

Fewer parameters means fewer numbers a domain expert has to invent.

Frame parameter reduction as an elicitation problem, not just a storage one: an expert can estimate 4 numbers reliably and 64 numbers not at all.

# Modelling Technique: Noisy OR

When many causes  $x_1, \dots, x_k$  can each independently trigger an effect  $y$ , a **noisy OR** models the effect as "each cause fires with its own probability, and the effect occurs if any of them fires":

$$p(y = 0 \mid x_1, \dots, x_k) = \prod_{i: x_i=1} (1 - \lambda_i)$$

$k$  parameters instead of  $2^k$ .

## Typical use

Disease–symptom networks, where many diseases give rise to the same symptom.

Ask what assumption buys the reduction: causes act independently and there is no interaction term. Then ask when that fails — synergistic causes — so the model choice stays a modelling decision.

# Uncertain and Unreliable Evidence

## Uncertain (soft) evidence

The state of  $x$  is itself uncertain, expressed as a distribution  $\tilde{x}$ . Infer by summing over the states:

$$p(y \mid \tilde{x}) = \sum_x p(y \mid x)p(x \mid \tilde{x})$$

*We look at the grass without touching it:*

$$p(T=1 \mid \tilde{T}) = 0.8.$$

## Unreliable evidence

The evidence is not trusted and is **replaced** by our own estimate:

$$p(y \mid x) \rightarrow p(h \mid x)$$

*Jack says his grass is wet, but Jack is in a bad mood and did not check.*

Both are drawn with a **dashed** edge.

The distinction is subtle and worth a moment: uncertain evidence keeps the observation and softens it unreliable evidence discards it and substitutes a judgement. Students who conflate them will model rumours as noise.

# Takeaway

A belief network is a **DAG plus one conditional table per node**. Modelling techniques — divorcing, noisy OR, soft evidence — all trade assumptions for parameters, and every trade should be stated out loud.

# Module 2

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## Reading Independence off the Graph

*Guiding Question: Two causes are independent until you observe their shared effect.  
Which pictures make that visible?*

# Three Ways Three Nodes Can Connect

Every conditional-independence question in a DAG reduces to these three patterns.

## Chain

$$a \rightarrow c \rightarrow b$$

$a$  and  $b$  are dependent — but **independent given  $c$** .

*Information flows along the chain until the middle is observed.*

## Common cause (fork)

$$a \leftarrow c \rightarrow b$$

$a$  and  $b$  are dependent — but **independent given  $c$** .

*The shared cause explains the correlation.*

## Common effect (collider)

$$a \rightarrow c \leftarrow b$$

$a$  and  $b$  are **independent** — but **dependent given  $c$** .

*Explaining away — the reverse of the other two.*

The whole slide is one table students should be able to reproduce. Emphasise the collider's inversion: conditioning normally destroys dependence, here it creates it. The wet-grass example is exactly a collider at T.

# Mini-Exercise

For each pair, decide whether observing the middle variable makes the outer two *more* or *less* related:

1. rain  $\rightarrow$  wet grass  $\rightarrow$  slippery pavement, observing wet grass
2. temperature  $\rightarrow$  ice-cream sales, temperature  $\rightarrow$  drownings, observing temperature
3. rain  $\rightarrow$  wet grass  $\leftarrow$  sprinkler, observing wet grass

# Mini-Exercise — Solution

Solution fragmented per the animation policy. Case 3 is the one that surprises; if the group is quick, ask what happens when you condition on a *descendant* of the collider (the same dependence appears — which is why d-separation mentions descendants).

# Mini-Exercise — Solution

## 1. Chain

Independent given wet grass  
— the pavement only cares  
that the ground is wet.

## 2. Fork

Independent given  
temperature — the classic  
spurious correlation dissolves.

## 3. Collider

*Dependent* given wet grass —  
learning it rained lowers belief  
in the sprinkler.

# d-Separation

A path between  $X$  and  $Y$  is **blocked** by a set  $Z$  if it contains either

- a chain or fork whose middle node **is in**  $Z$ , or
- a collider whose middle node — and all of its descendants — are **not in**  $Z$ .

If every path from  $X$  to  $Y$  is blocked by  $Z$ , then  $X$  and  $Y$  are **d-separated** given  $Z$ , and  $X \perp \perp Y \mid Z$ .

d-separation is sound: what it declares independent really is independent for every distribution factorising over the graph. It is not complete for a *particular* distribution — extra independences may hold numerically without being visible in the graph.

Definition after the three cases, so it reads as a summary rather than an axiom. The soundness/completeness caveat is the honest version of "graphs have limited expressive power" from the intro unit.

# Takeaway

Independence in a belief network is a **graph property**: chains and forks are blocked *by* observing the middle node, colliders are opened by it. d-separation is that rule applied to every path.

# Module 3

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## Graphical Notation

***Guiding Question:*** *A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn?*

# Drawing a Model

A probabilistic model is a factorisation, and the graph is its picture. Four conventions carry almost all the information:

## Shaded node

An **observed** variable — its value is in the data.

## Unshaded node

A **latent** variable or a parameter — inferred, never seen.

## Small solid dot

A fixed **hyperparameter**, not inferred.

## Arrow

A conditional dependence: the child's distribution is conditioned on the parent.

Insist that this is notation, not new theory: the diagram is a compact way to write a factorisation students can already read. The shading convention is what they will most often need when reading papers.

# Plates

A **plate** is a rectangle drawn around part of the graph with a count in its corner: *repeat this structure  $N$  times, with independent copies of the variables inside.*

It is the i.i.d. assumption, drawn. Without it, a model of  $N$  observations would need  $N$  copies of the same subgraph.

**Nested plates** express grouped data —  $N$  documents each of  $M_n$  words, or  $K$  hospitals each with  $N_k$  patients.

## What sits outside the plate

Anything **shared** across observations — the parameters. That is precisely what learning estimates.

# Reading a Diagram

For a Bayesian linear regression: a hyperparameter dot feeds  $\mathbf{w}$ , which sits **outside** the plate inside a plate over  $N$  observations sit  $\mathbf{x}_i$  (shaded) and  $y_i$  (shaded), with  $\mathbf{w}$  pointing into each  $y_i$ .

Read off the factorisation directly:  $p(\mathbf{w} \mid \tau) \prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w})$ . The plate became the product the arrows became the conditioning bars.

# Mini-Exercise

A diagram shows a dot  $\rightarrow \alpha \rightarrow \theta$  inside a plate of  $K$ , and  $\theta \rightarrow z_i \rightarrow x_i$  inside a plate of  $N$ , with  $x_i$  shaded.

1. Which variables are inferred?
2. What does the  $N$  plate assert?
3. Write the factorisation.

# Mini-Exercise — Solution

Plate notation compresses a factorisation into a diagram: **shading** marks what was observed, **plates** mark what repeats independently, and **what sits outside the plates** is what learning has to estimate.

Part 3 is the skill worth having: converting a diagram into a formula and back. Every probabilistic-modelling paper the students read assumes it.

# Mini-Exercise — Solution

1.  $\theta$  and every  $z_i$  — unshaded. Only  $x_i$  is observed  $\alpha$  is fixed.

2. The  $N$  observations are **conditionally independent given  $\theta$**  — each with its own latent  $z_i$ .

3.  $p(\theta \mid \alpha) \prod_{i=1}^N p(z_i \mid \theta) p(x_i \mid z_i)$  — a mixture model.

Plate notation compresses a factorisation into a diagram: **shading** marks what was observed, **plates** mark what repeats independently, and **what sits outside the plates** is what learning has to estimate.

# Module 4

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## Undirected Models

**Guiding Question:** *Pixels influence each other symmetrically — no pixel causes its neighbour. How do we model that?*

# Undirected Models

**Markov network (Markov random field).** An **undirected** graph in which the joint distribution factorises over the **cliques**  $c \in \mathcal{C}$  of the graph:

$$p(x_1, \dots, x_n) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \phi_c(x_c), \quad Z = \sum_x \prod_c \phi_c(x_c)$$

The  $\phi_c \geq 0$  are **potentials** — arbitrary non-negative functions, not probabilities.  $Z$  is the normalising **partition function**.

When all cliques have size 2 the model is a **pairwise Markov network**.

The key difference from a BBN: potentials are not conditional probabilities and do not normalise locally, so a global  $Z$  is required. Computing  $Z$  is where the cost lives.

# Directed vs. Undirected

## Belief network

Directed, acyclic. Factors are conditional probabilities. Natural for **causal or generative** stories conditioning on a collider creates dependence.

## Markov network

Undirected. Factors are potentials with a global  $Z$ . Natural for **symmetric interactions** — pixels, atoms, neighbours in a lattice.

Neither subsumes the other: some independence patterns are expressible only in one. Give the standard examples if time permits (a collider is not expressible undirected a 4-cycle is not expressible directed).

# Marginalising and Conditioning Behave Oppositely

## Marginalising over $C$

makes  $A$  and  $B$  **dependent** — the sum over all states of  $c$  couples the potentials.

## Conditioning on $C$

In the chain  $A - C - B$ , a fixed  $c$  is a constant in the potentials, so the factor splits and  $A \perp\!\!\!\perp B \mid C$ .

*This holds because  $C$  separates them — not for every  $C$ .*

This pair is the undirected counterpart to chains, forks and colliders. In an undirected graph the rule is simple: separation *is* conditional independence, no collider exception.

# Markov Properties and the Separation Test

For undirected models, independence is plain graph separation: if every path from  $X$  to  $Y$  passes through  $Z$ , then  $X \perp\!\!\!\perp Y \mid Z$ .

For a **directed** model the same question is answered by converting the graph:

## 1. Ancestral graph

Keep  $X \cup Y \cup Z$  and their ancestors delete all other nodes and their edges.

## 2. Moralisation

Connect any two nodes that share a child, then drop all arrow heads.

## 3. Separation

Remove the links neighbouring  $Z$ . If no path joins  $X$  to  $Y$ , then  $X \perp\!\!\!\perp Y \mid Z$ .

"Marrying the parents" is where moralisation gets its name, and it is exactly the collider rule made structural: parents of a common child become connected, so they are no longer separable. This procedure is equivalent to d-separation.

# Two Examples

## Boltzmann machine

A pairwise Markov network over binary units with weights on the edges the restricted variant (RBM) allows edges only between a visible and a hidden layer.

## Image cleanup

Pixels form a lattice: each potential prefers a pixel to agree with its neighbours and with the noisy observation. Inference denoises the image.

The image example is the most intuitive Markov network there is — smoothness as a potential. It also previews the energy-based view used for neural networks later.

# Other Graphical Models

## Chain graphs

Combine directed and undirected edges in one model.

## Factor graphs

Make each factor an explicit node — mostly used **for inference**, since they state precisely which variables each factor touches.

Undirected models trade causal direction for symmetric interaction, and local normalisation for a global partition function. Separation in the (moralised) graph is the single test for conditional independence.

# Module 5

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## Fitting and Inference

**Guiding Question:** *We can now write the model down. Can we actually compute anything with it?*

# Three Questions You Can Ask a Model

## **Infer unobserved variables**

Compute the posterior given evidence — probabilistic inference.

## **Learn the parameters**

Estimate the factors' distributions from data.

## **Learn the structure**

Discover which dependencies exist at all.

The three get progressively harder — and progressively more interesting.

Structure learning is where causal discovery lives flag it as an open research area rather than a solved technique, since students will meet confident claims about it.

# Inference Is Hard in General

Both exact and approximate inference in belief networks are **NP-hard**. Whether a particular model is tractable depends on its structure.

## Exact inference

Variable elimination, junction trees, message passing.  
Efficient when the graph is a tree or has low treewidth.

## Approximate inference

Sampling (MCMC, importance sampling) and variational methods.  
Trade a guarantee for tractability.

## Speaker notes

The practical rule: structure decides. A chain or tree is linear a dense graph is hopeless. This is why modelling and inference are not separable concerns — a model you cannot query is not a model.



# Learning the Parameters

With **fully observed** data, parameter learning is counting: each conditional table is estimated from the frequencies of its variable and parents.

With **hidden variables**, the counts are unavailable and one alternates between inferring the hidden values and re-estimating the parameters (expectation–maximisation).

The estimates are only as good as the sample supporting each table row. A parent combination that never occurs in the data leaves its row undetermined — which is another argument for keeping tables small.

Connect back to divorcing and noisy OR: parameter reduction is also a statistical necessity, not only a storage convenience. This also foreshadows smoothing in naive Bayes.

# Takeaway

Graphical models make the joint distribution **writable** whether they make it **queryable** depends on the graph's structure. Parameters come from counting where data is complete, and from iterating where it is not.